Software Requirements Specification

for

Library Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This document provides a detailed description of the requirements for the Library Management System (LMS). It defines both functional and non-functional requirements of the system, covering all major operations such as user validation, book search, availability checking, issuing/reserving books, returning books, fine calculation, reporting lost books, and database updates.

1.2 Document Conventions

This document follows a structured format based on standard SRS guidelines. Requirements are organized into sections and uniquely identified. Functional requirements describe system behavior, while non-functional requirements define performance and quality attributes

1.3 Intended Audience and Reading Suggestions

This document is intended for developers, testers, project managers, and end users. Developers will use it for implementation, testers for validation, and users for understanding system functionality.

1.4 Product Scope

The Library Management System is designed to automate and manage library operations efficiently. It reduces manual effort, ensures accuracy, and provides quick access to book-related information.

1.5 References

- IEEE Software Requirements Specification Standards
- Software Engineering Guidelines
- Library Management System Design Notes

2. Overall Description

2.1 Product Perspective

The Library Management System is a standalone application that interacts with users and a central database. The system ensures that only valid users can access services. Once authenticated, users can search for books, check availability, and perform issue/return operations.

2.2 Product Functions

The Library Management System performs the following major functions:

- User authentication
- Book search
- Availability checking
- Book issuing and reservation
- Book return processing
- Fine calculation
- Lost book reporting
- Database management

2.3 User Classes and Characteristics

User Class 1: Library User

Searches, issues, and returns books.

User Class 2: Library Database System

Stores and manages all records related to books and users.

2.4 Operating Environment

The Library Management System operates in a desktop environment.
Operating System: Windows or Linux
Hardware Platform: Personal computer
Memory: Minimum 4GB RAM
Database: SQL-based database system
Input Devices: Keyboard and mouse
Output Device: Monitor

2.5 Design and Implementation Constraints

The following constraints apply to the system:

- Only authenticated users can access the system
- Each book must have a unique ID
- Database must be updated after every transaction

2.6 User Documentation

The Library Management System will include the following user documentation:

- User manual describing system usage
- Help screen explaining system operations
- Error message descriptions
- Instructions for issuing and returning books

2.7 Assumptions and Dependencies

The following assumptions are made:

- Users are pre-registered
- Books are available in the database
- Fine rules are predefined

3. External Interface Requirements

3.1 User Interfaces

The Library Management System provides user interfaces for login, book search, availability display, issue or reserve book, return book, fine display, and lost book reporting. The user interface includes input fields for user ID and book details. The system displays availability status, issue confirmation, fine amount, and return status.

3.2 Hardware Interfaces

The system interacts with standard computer hardware components including keyboard for input, mouse for selection, and monitor for displaying system output.

3.3 Software Interfaces

The Library Management System interacts with a database system that stores user details and book information. The system also interacts with the operating system for execution. Input data includes user ID, book ID, and return date. Output data includes availability status, issue confirmation, fine details, and updated book status.

3.4 Communications Interfaces

The Library Management System does not require external communication interfaces. The system operates as a standalone application. No network communication protocols are required.

4. System Features

4.1 User Validation

4.1.1 Description and Priority

The system verifies user credentials before granting access. Invalid users are denied access.

4.1.2 Stimulus/Response Sequences

The user enters login credentials. The system verifies the credentials. If the user is valid, the system allows further operations. If the user is invalid, access is denied.

4.1.3 Functional Requirements

- Accept login credentials
- Validate user identity
- Allow access for valid users
- Reject invalid users

4.2 Search Book

4.2.1 Description

Users can search for books using various parameters such as title, author, or ID. The system retrieves matching results.

4.2.2 Functional Requirements

- Accept search input
- Query database
- Display results

4.3 Check Availability

4.3.1 Description

The system checks whether the selected book is available and displays the status.

4.3.2 Functional Requirements

- Verify availability
- Show available/unavailable status

4.4 Issue / Reserve Book

4.4.1 Description

If available, the system allows issuing or reserving the book and assigns a due date.

4.4.2 Functional Requirements

- Issue book
- Reserve book
- Assign due date
- Update database

4.5 Return Book

4.5.1 Description

The system processes returned books and checks whether they are returned on time or late.

4.5.2 Functional Requirements

- Accept returned book
- Compare return date with due date
- Update records

4.6 Fine Calculation

4.6.1 Description

If the return is delayed, the system calculates and displays the fine.

4.6.2 Functional Requirements

- Calculate fine
- Display fine amount

4.7 Report Lost Book

4.7.1 Description

Users can report lost books, and the system updates records accordingly

4.7.2 Functional Requirements

- Mark book as lost
- Update database

5. Nonfunctional Requirements

5.1 Performance Requirements

The system shall respond to user input within two seconds. Book search and availability checking shall be performed quickly. Database updates shall be completed immediately after transactions.

5.2 Security Requirements

The system shall authenticate users before allowing access. Only authorized users shall issue or return books. The system shall protect database information.

5.3 Reliability

The system shall be reliable and easy to use. The system shall be maintainable and efficient. The system shall provide accurate results. The system shall be robust against invalid inputs.

5.4 Business Rules

Only valid users can issue books. Books must be available before issuing. Fine shall be charged for late returns. Lost books must be reported. Database must be updated after each transaction.

6. Other Requirements

The system shall maintain database tables for users, books, issue records, return records, and fine details. The system shall store issue date, due date, and return date.

Appendix A: Glossary

LMS – Library Management System
SRS – Software Requirements Specification

Appendix B: Analysis Models

System Flow:

- User logs in
- System validates user
- User searches for a book
- System checks availability
- Book is issued/reserved if available
- User returns the book
- System determines:
    - On-time return
    - Late return (fine applied)
    - Lost book
- Database is updated
-

Appendix C: To Be Determined List

- User records
- Book records
- Issue/return logs
- Fine details

Appendix D:Future Enhancements

- Online access
- Mobile application support
- Barcode integration